

Mei-Lin Chow

Senior ML Engineer | Fintech AI Systems

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PROFESSIONAL SUMMARY

ML engineer specializing in LLM applications for financial services. Expertise in retrieval-augmented generation, multi-agent orchestration, and fine-tuning workflows. Track record building sophisticated AI systems that integrate with complex domain APIs and payment infrastructure.

EXPERIENCE

Senior ML Engineer | Vertex Financial Technologies | *January 2024 – Present*

- Architected RAG pipeline for transaction classification leveraging vector embeddings and semantic search; achieved 94.7% F1 on internal benchmark dataset of 50K labeled transactions
- Designed multi-agent system for payment reconciliation using Claude and GPT-4; agent successfully routes queries across three independent financial APIs with 99.2% accuracy on evaluation set
- Implemented fine-tuning workflow for merchant categorization model; improved baseline accuracy from 87% to 92.3% using LoRA on 100K curated financial transaction records
- Open-sourced fintech-RAG library (2.1K GitHub stars); published technical blog post on LLM evaluation methodologies that reached 8K readers
- Collaborated with product and compliance teams to define eval frameworks for model safety in payment contexts

ML Engineer | Cascade Payment Systems | *March 2023 – December 2023*

- Built agentic system for fraud detection featuring dynamic tool use; evaluated against 500K historical transaction patterns with latency under 200ms per inference
- Developed custom fine-tuning pipeline for dispute resolution; trained 7B parameter model achieving 91% accuracy on hold-out test set of 5K disputes
- Integrated embedding-based similarity search into payment routing system; constructed benchmark showing 15% improvement over heuristic baseline on simulated transaction flows
- Presented technical talk on LLM evaluation at FinTech AI meetup (85 attendees)

Machine Learning Intern | Northbridge Capital | *June 2022 – February 2023*

- Prototyped agentic workflows for portfolio analysis using multi-turn API interactions; benchmarked on 10 market scenarios

SKILLS

Python, PyTorch, LangChain, Claude API, OpenAI API, Hugging Face, Vector Databases, RAG, Fine-tuning, Multi-agent systems, Financial APIs